

1. (20 marks) For $(a, b) \in \mathbb{R}^2$, consider the following quadratic form

$$Q(\mathbf{x}) = a(x_1^2 + x_3^2) + bx_2^2 + 2a(x_1x_2 + x_2x_3).$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is, find the symmetric matrix A satisfying $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$. Then, find all the principal minors of A .
- (b) (12 marks) Define what it means for A to be *indefinite*, *positive semidefinite* and *negative semidefinite*. Then, find all parameter values for a and b such that the quadratic form $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite*?
- (c) (4 marks) Plot the pairs (a, b) for which $Q(\mathbf{x})$ is respectively *indefinite*, *positive semidefinite* and *negative semidefinite* in $\mathbb{R}^2$.
2. (20 marks) Consider the functions $f(\mathbf{x}) = -x_1^2 x_2^2$ defined on some convex set U in $\mathbb{R}^2$.
- (a) (4 marks) Define what it means for f to be *quasi-concave* and *quasi-convex*.
- (b) (16 marks) Is f *quasi-concave* or *quasi-convex* when U is equal to $\mathbb{R}^2$ and $\mathbb{R}_+^2$ respectively? You may exploit results proven in lecture, and use counterexamples to show that a property fails for some U .
3. (40 marks) For $a > 0$, consider the problem:

$$\max_{x,y} ax + y \quad \text{s.t.} \quad x^2 + y^2 \leq 3/4, \quad x \geq 0, \quad y \geq a.$$

- (a) (7 marks) Is the objective function concave? Are the constraints convex? For what values of $a > 0$ is Slater condition satisfied? So, for what values of $a > 0$, can you solve this optimization problem using the *Kuhn-Tucker method*?
- (b) (7 marks) Suppose the restrictions on a found in part (a) hold. Write down the *Kuhn-Tucker necessary first order conditions* that must be satisfied by the solution of the constrained optimization problem. Are solutions to these conditions also maximizers of the Lagrangian?
- (c) (20 marks) Solve the constrained optimization problem for all values of $a > 0$ satisfying the restrictions on a found in part (a).
- (d) (6 marks) Suppose the restrictions on a found in part (a) hold. Use the Envelope Theorem to express the slope of value function with respect to the parameter a in terms of the solution $(x(a), y(a), \boldsymbol{\lambda}(a))$ found in part (c).
4. (20 Marks) Consider the correspondence $\mathbf{h} : [0, 2]^2 \rightrightarrows [0, 2]^2$, where $\mathbf{h}(\mathbf{x}) = (h_1(\mathbf{x}), h_2(\mathbf{x}))$ and

$$h_1(\mathbf{x}) = \begin{cases} 2 & \text{if } x_2 \in (1/2, 3/2) \\ [0, 2] & \text{if } x_2 \in \{1/2, 3/2\} \\ 0 & \text{if } x_2 \in [0, 1/2) \cup (3/2, 2] \end{cases} \quad \text{and} \quad h_2(\mathbf{x}) = \begin{cases} [x_1^2/2, x_1^{0.5}/2] & \text{if } x_1 \in [0, 1] \\ [x_1^{0.5}/2, x_1^2/2] & \text{if } x_1 \in [1, 2] \end{cases}.$$

Argue why *Kakutani fixed point theorem* applies. Then, find all the *fixed points* of this correspondence.